Arista Access Node MAR-ACC-49 Marathahalli Core QoS Congestion Event Resolution SOP

Node: MAR-ACC-49 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3361

Description

This SOP aims to address a QoS congestion event on the Arista Access Node MAR-ACC-49 located at the Marathahalli Core site. The failure domain is primarily attributed to microbursts and interface tail drops, leading to QoS queue starvation. The underlying protocol responsible for this phenomenon is the IEEE 802.1p priority-based queuing mechanism and the associated packet scheduling algorithms. The hardware component, in this case, is the egress interface of the Arista Access Node. The QoS congestion event is a result of the interface's inability to handle the incoming packet rate, leading to packet drops and subsequent queue starvation. The probability of this event occurring can be attributed to various factors, including suboptimal interface configuration, inadequate buffer sizes, or excessive packet overhead.

Pre-requisite Checks

- Verify that the Marathahalli Core site is accessible and the remote hands are dispatched to the location.

- Check the status of the redundant link to ensure that it is operational and configured correctly.

- Review the network topology to identify any potential bottlenecks or areas of congestion that may be contributing to the QoS congestion event.

Potential Root Causes

Cause	Probability
Suboptimal interface configuration	High
Inadequate buffer sizes	Medium
Excessive packet overhead	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface egress | include drops, Expected Output: The output should indicate the total number of packets dropped on the egress interface., Description: This command isolates the egress interface's packet drop statistics, providing insight into the interface's ability to handle incoming packets.

Command: show qos queue | include drops, Expected Output: The output should indicate the number of packets dropped in the specified QoS queue., Description: This command isolates the QoS queue's packet drop statistics, providing insight into the queue's ability to handle incoming packets.
```

Resolution Steps

Step 1: Step: Verify Interface Configuration, Command: show interface egress | include configuration, Description: This step involves verifying the interface configuration to ensure that it is set up correctly to handle the incoming packet rate. Check for any suboptimal settings that may be contributing to the QoS congestion event.

Step 2: Step: Adjust Buffer Sizes, Command: config interface egress buffer-size , Description: This step involves adjusting the buffer sizes on the egress interface to ensure that it can handle the incoming packet rate without dropping packets. The buffer size should be set to a value that is optimal for the interface's configuration and the network's requirements.

Step 3: Step: Optimize Packet Overhead, Command: config qos queue overhead , Description: This step involves optimizing the packet overhead on the specified QoS queue to ensure that it can handle incoming packets without dropping them. The packet overhead should be set to a value that is optimal for the queue's configuration and the network's requirements.

Post-Check Verification

```
Command: show interface egress | include drops, Expected Output: The output should indicate zero packets dropped on the egress interface., Description: This command verifies that the egress interface is able to handle incoming packets without dropping them.

Command: show qos queue | include drops, Expected Output: The output should indicate zero packets dropped in the specified QoS queue., Description: This command verifies that the QoS queue is able to handle incoming packets without dropping them.
```

Roll-back Procedures

- Step: Restore Original Interface Configuration, Command: config interface egress restore, Description: This step involves restoring the original interface configuration to ensure that the interface is set up correctly to handle incoming packets.

- Step: Revert Buffer Sizes, Command: config interface egress buffer-size , Description: This step involves reverting the buffer sizes on the egress interface to their original values to ensure that the interface can handle incoming packets without dropping them.

- Step: Revert Packet Overhead, Command: config qos queue overhead , Description: This step involves reverting the packet overhead on the specified QoS queue to its original value to ensure that the queue can handle incoming packets without dropping them.